MIL-C-55446A
19 February 1968

SUPERSEDING
MIL-C-55446(EL)
29 November 1965

MILITARY SPECIFICATION

CABLES, TELEPHONE, SWITCHBOARD: PLASTIC INSULATED, PLASTIC JACKETED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plastic insulated, plastic jacketed telephone switchboard cables consisting of No. 22 AWG conductors in combinations of pairs and singles ranging from 4 pairs to 201 pairs. (See 1.2.)

1.2 Classification. The following switchboard cables are covered by this specification.

No. of pairs and singles	Total No. of conductors	Maximum average overall diameters of finished cables
No. 22 AWG		Inches
4 pairs	8	0.360
5 pairs, 5 singles	15	0.440
6 pairs	12	0.390
8 pairs, 8 singles	24	0.480
10 pairs, 10 singles	30	0.540
11 pairs	22	0.470
16 pairs	32	0.530
20 pairs	40	0.560
20 pairs, 10 singles	50	0.610
20 pairs, 20 singles	60	0.660
25 pairs	50	0.610
26 pairs, 26 singles	78	0.720
30 pairs	60	0.640
40 pairs	80	0.720
41 pairs, 41 singles	123	0.830
51 pairs	102	0.800
76 pairs	152	0.910
101 pairs	202	1.030
101 pairs, 100 singles	302	1.250
201 pairs	402	1.410

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-343	Wire, Electrical (Uninsulated).

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-I-3930	Insulating and Jacketing Compounds, Electrical (For Cables, Cords and Wires).
MIL-C-12000	Cable, Cord, and Wire, Electric; Packaging of.

STANDARDS

FEDERAL

FED-TEST-STD-228	Cable and Wire, Insulated; Methods of Testing.

MILITARY

MIL-STD-104	Limits for Electrical Insulation Color.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. The cables covered by this specification are switchboard cables, the conductors of which are No. 22 AWG solid copper wire, tinned, and individually insulated with an extruded polyvinyl chloride covering. The insulation is color coded for convenient identification of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

pairs and single wires. The core is made up of pairs and single wires as required, and cabled either concentrically in layers or in unit construction, depending on its size. The individual units of unit constructed cables are bound with color coded spaced servings of yarn or polyethylene tape binders. The core is covered with an overlapped spiral winding of paper or, at the option of the manufacturer, polyester tape. Over this paper or polyester covering and beneath the outer sheath is a nylon rip cord to facilitate stripping. The outer sheath is an extruded polyvinyl chloride jacket.

3.2 Construction. The cable shall be made up of the number of pairs and single wires as required. The pairs shall be twisted, with the maximum length of pair twist not to exceed 6 inches. To minimize cross talk, each pair in a concentric type cable and each pair in a unit of unit construction shall have a twist with a length of lay different from that of any other pair in the cable or unit respectively.

3.3 Materials. The materials for the cables covered by this specification shall be as specified herein. When a definite material is not designated, the best material commercially available for the purpose shall be used.

3.4 Conductors. Each conductor of the cables shall be No. 22 AWG solid, soft or drawn and annealed tinned copper wire in accordance with Specification QQ-W-343.

3.4.1 Joints in conductors. If joints are necessary, they shall be made by brazing with silver solder or by welding. The diameter of the wire measured at the joint shall be within the diameter tolerance for the conductors. The conductor surface at the joint shall be smooth and free from lumps and sharp projections. The tensile strength of a section of the conductor containing a joint shall be not less than 90 percent of the tensile strength of an adjacent section of the conductor of equal length free from joints. The resistance of a section of the conductor not exceeding 6 inches in length, which includes the joint, shall be not more than 105 percent of the resistance of an adjacent section of equal length free from joints.

3.5 Insulation. Each conductor of the cables shall be insulated with an extruded layer of semi-rigid polyvinyl chloride. The polyvinyl chloride compound shall be in accordance with Specification MIL-I-3930, type IP. When the requirements of MIL-I-3930 are in conflict with 3.10 of this specification, the latter shall apply.

3.5.1 Insulation thickness. The extruded layer of insulating compound applied to the conductor shall be not less than 0.008 inch thick.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

3.5.2 Diameter of insulated conductor. The diameter over the insulated conductor shall be not more than 0.048 inch.

3.5.3 Color code. The insulated conductors shall be color coded so as to readily identify pairs and single conductors. The color code shall be in accordance with Table I for pairs and Table II for singles.

Table I - Color code for pairs.

Pair No.	Wire		Mate	
	Wide Stripe (Base Color)	Narrow Stripe	Narrow Stripe	Wide Stripe (Base Color)
P 1	Blue	White	Blue	White
P 2	Orange	White	Orange	White
P 3	Green	White	Green	White
P 4	Brown	White	Brown	White
P 5	Slate	White	Slate	White
P 6	Blue	Red	Blue	Red
P 7	Orange	Red	Orange	Red
P 8	Green	Red	Green	Red
P 9	Brown	Red	Brown	Red
P 10	Slate	Red	Slate	Red
P 11	Blue	Black	Blue	Black
P 12	Orange	Black	Orange	Black
P 13	Green	Black	Green	Black
P 14	Brown	Black	Brown	Black
P 15	Slate	Black	Slate	Black
P 16	Blue	Yellow	Blue	Yellow
P 17	Orange	Yellow	Orange	Yellow
P 18	Green	Yellow	Green	Yellow
P 19	Brown	Yellow	Brown	Yellow
P 20	Slate	Yellow	Slate	Yellow
Spare Pair:				
SP 1	White		Red	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

Table II - Color code for singles.

Single No.	Color Coding		
	Base Color	Stripe (wide)	Stripe (narrow)
S 1	Blue		
S 2	Orange		
S 3	Green		
S 4	Brown		
S 5	Slate		
S 6	Blue	Red	White
S 7	Orange	Red	White
S 8	Green	Red	White
S 9	Brown	Red	White
S 10	Slate	Red	White
S 11	Blue	Black	White
S 12	Orange	Black	White
S 13	Green	Black	White
S 14	Brown	Black	White
S 15	Slate	Black	White
S 16	Blue	Yellow	White
S 17	Orange	Yellow	White
S 18	Green	Yellow	White
S 19	Brown	Yellow	White
S 20	Slate	Yellow	White
Spare Single: SS 1	Red	White	

3.5.3.1 Insulation color. The insulation color, including spiral colors, shall be within the limits specified in MIL-STD-104. The base color shall be integral. The color striping shall be either integral or applied to the insulation by any commercially accepted standard method, provided the colors are readily distinguishable on the insulation after cable has been subjected to the tests of the specification, except for the flame retardation tests. The colors shall be durable, and shall not abrade as a result of cable handling or abrasion.

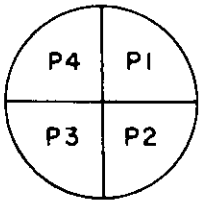
3.5.3.2 Color stripe length of lay. The color stripe length of lay shall not exceed 1 inch.

3.6 Cabling. The required number of pairs and single conductors shall be cabled in layers as a concentric-type construction or in units as a unit-type construction, as shown in Figures 1 through 6.

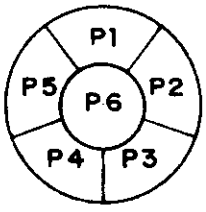
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

FIGURE 1. CABLE CONFIGURATION-LAYER TYPE CONSTRUCTION

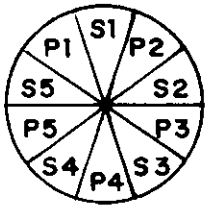
8 CONDUCTORS (4 PAIRS)



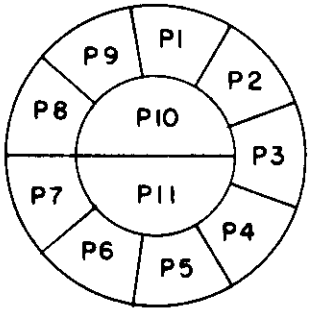
12 CONDUCTORS (6 PAIRS)



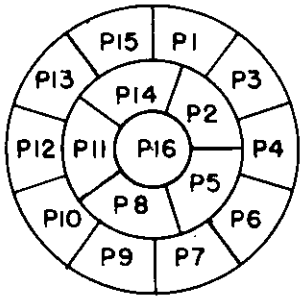
15 CONDUCTORS (5 PAIRS, 5 SINGLES)



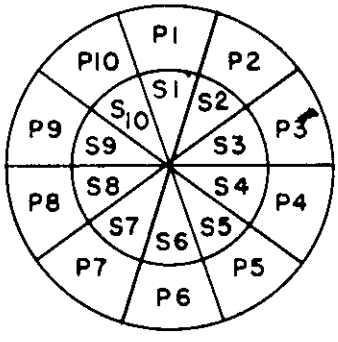
22 CONDUCTORS (11 PAIRS)



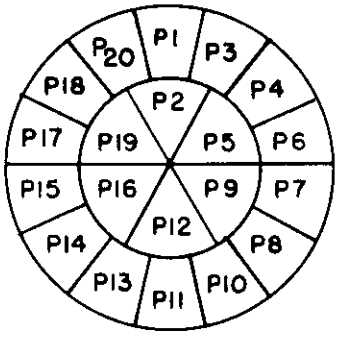
32 CONDUCTORS (16 PAIRS)



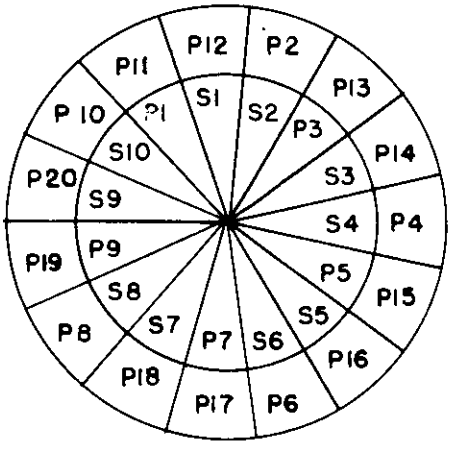
30 CONDUCTORS (10 PAIRS 10 SINGLES)



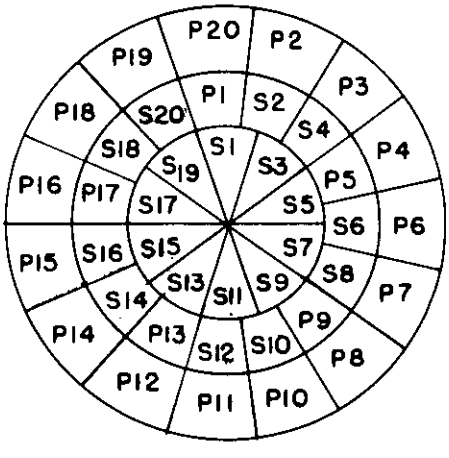
40 CONDUCTORS (20 PAIRS)



50 CONDUCTORS (20 PAIRS 10 SINGLES)



60 CONDUCTORS (20 PAIRS 20 SINGLES)



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

FIGURE 2 CABLE CONFIGURATION

FIGURE 2a LAYER TYPE CONSTRUCTION

24 CONDUCTORS (8 PAIRS - 8 SINGLES)

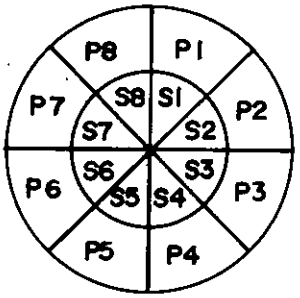
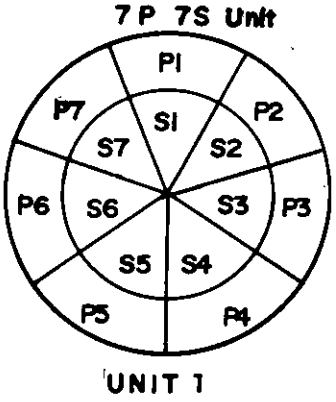
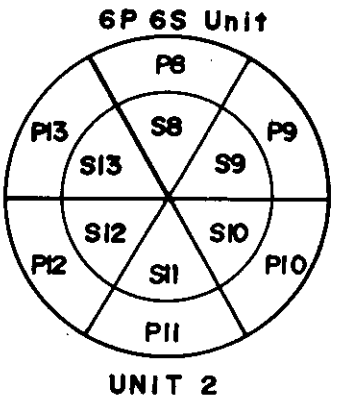
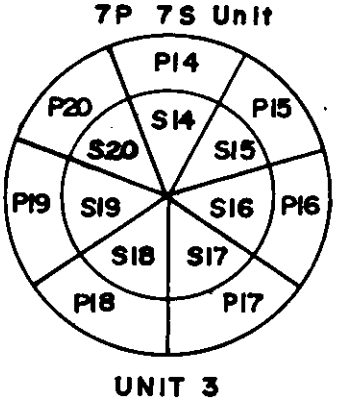


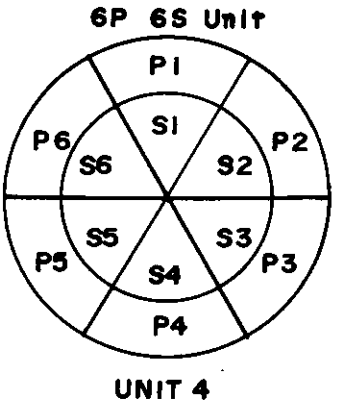
FIGURE 2b UNIT TYPE CONSTRUCTION

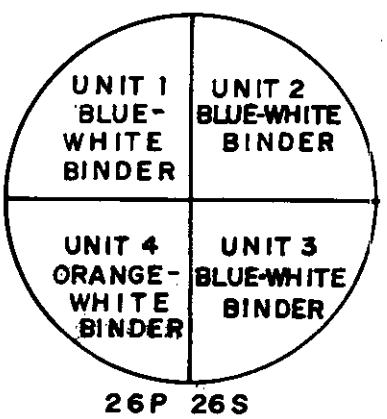
78 CONDUCTORS (26 PAIRS - 26 SINGLES)







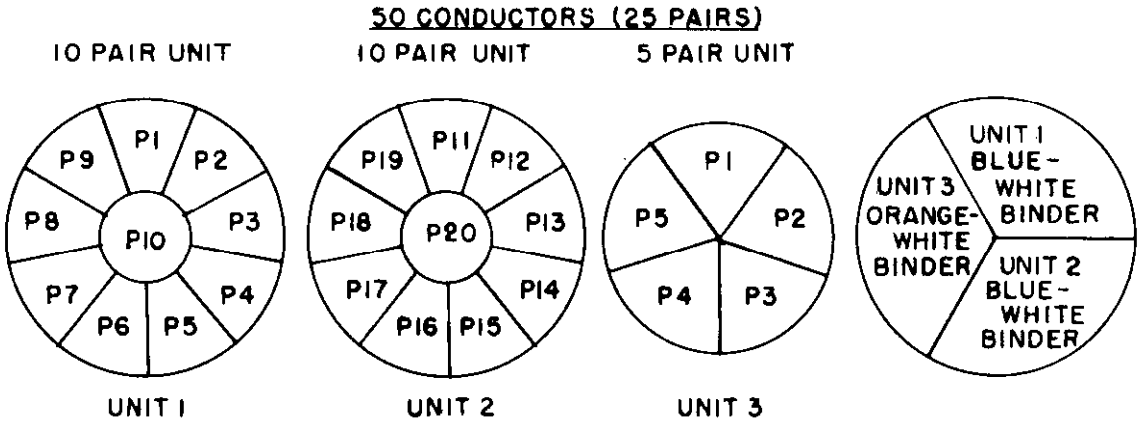




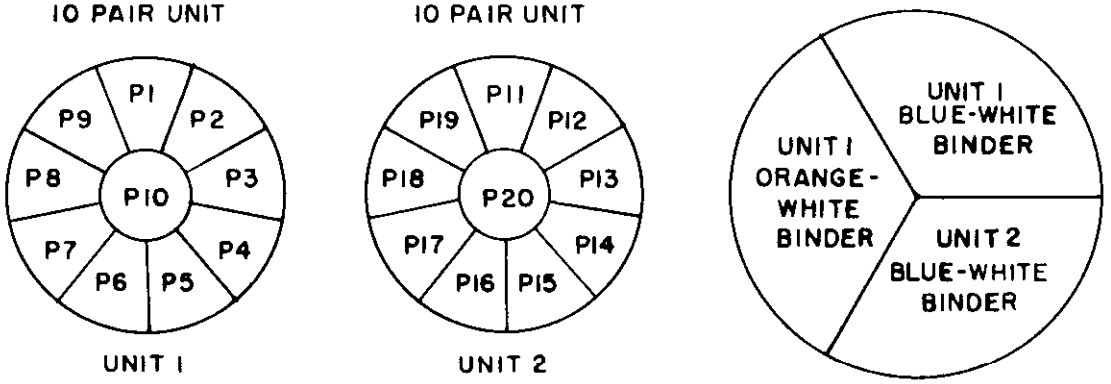
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

FIGURE 3. CABLE CONFIGURATION - UNIT TYPE CONSTRUCTION

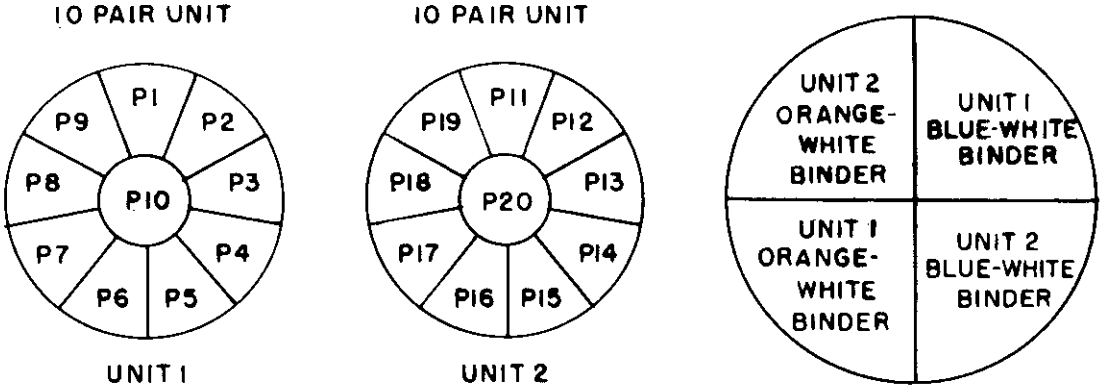
50 CONDUCTORS (25 PAIRS)



60 CONDUCTORS (30 PAIRS)



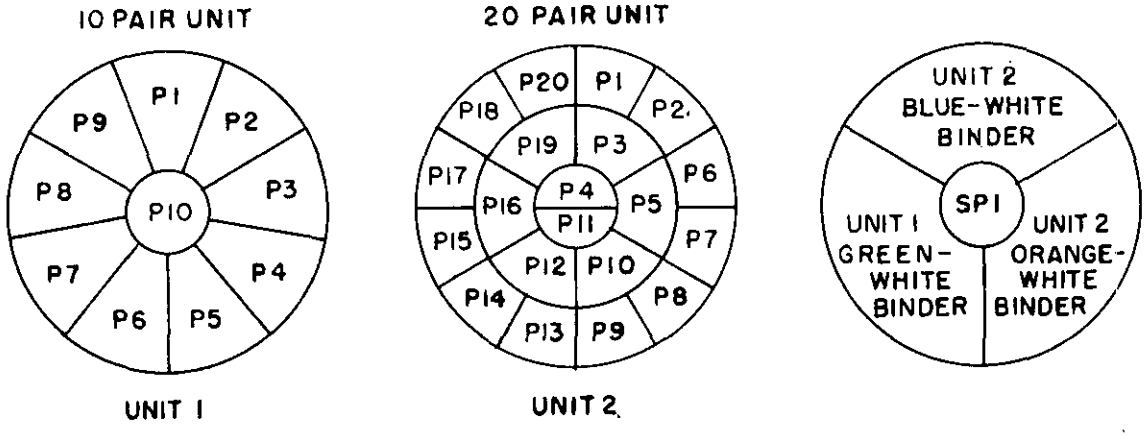
80 CONDUCTORS (40 PAIRS)



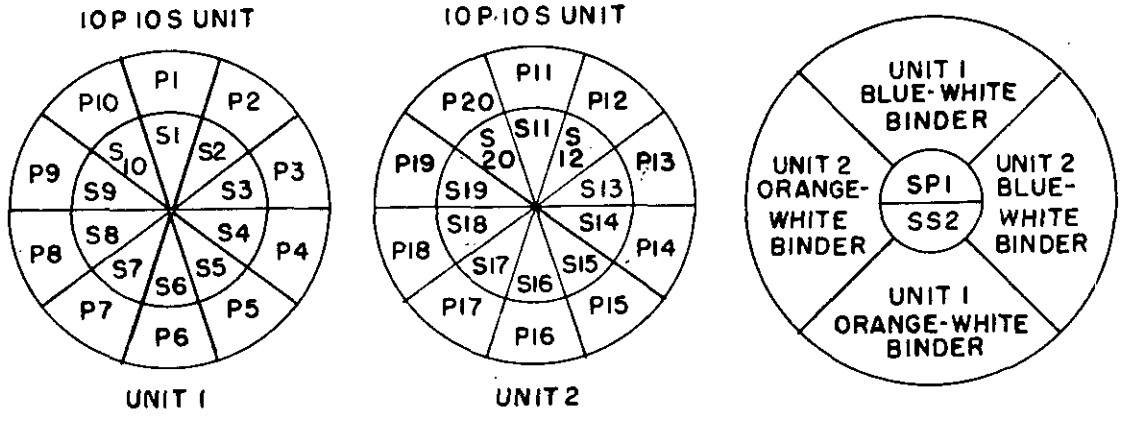
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

FIGURE 4. CABLE CONFIGURATION-UNIT TYPE CONSTRUCTION

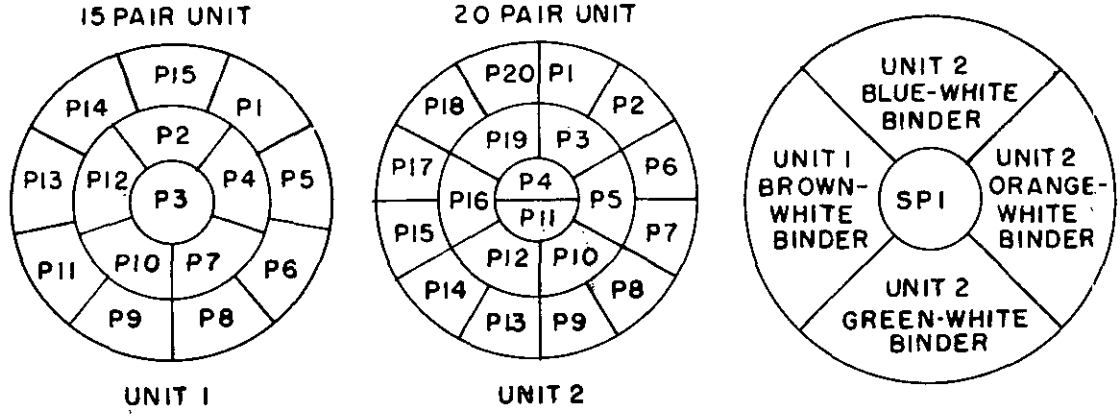
102 CONDUCTORS (50 PAIRS - 1 SINGLE PAIR)



123 CONDUCTORS (40 PAIRS-40 SINGLES, WITH 1 SPARE PAIR AND 1 SPARE SINGLE)



152 CONDUCTORS (75 PAIRS WITH 1 SPARE PAIR)

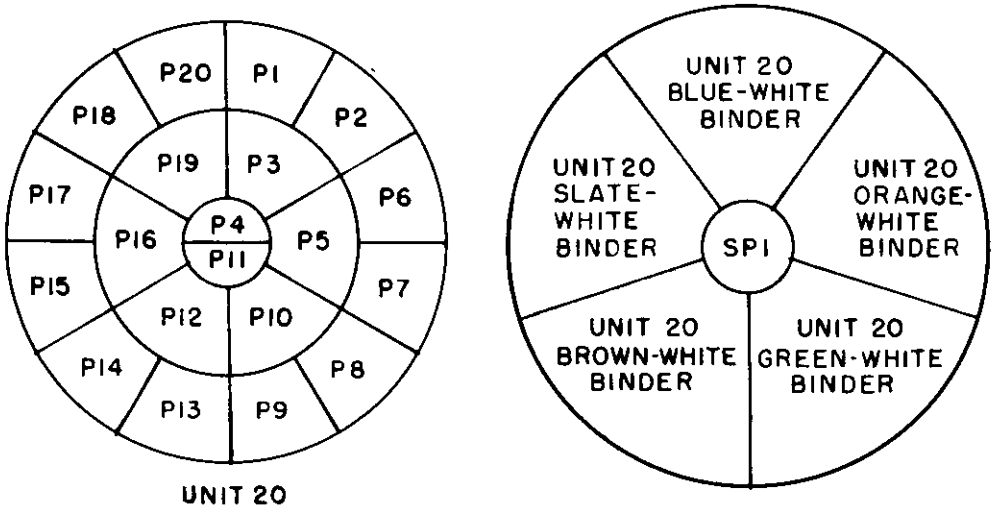


Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

FIGURE 5. CABLE CONFIGURATION-UNIT TYPE CONSTRUCTION

202 CONDUCTORS (100 PAIRS WITH 1 SPARE PAIR)

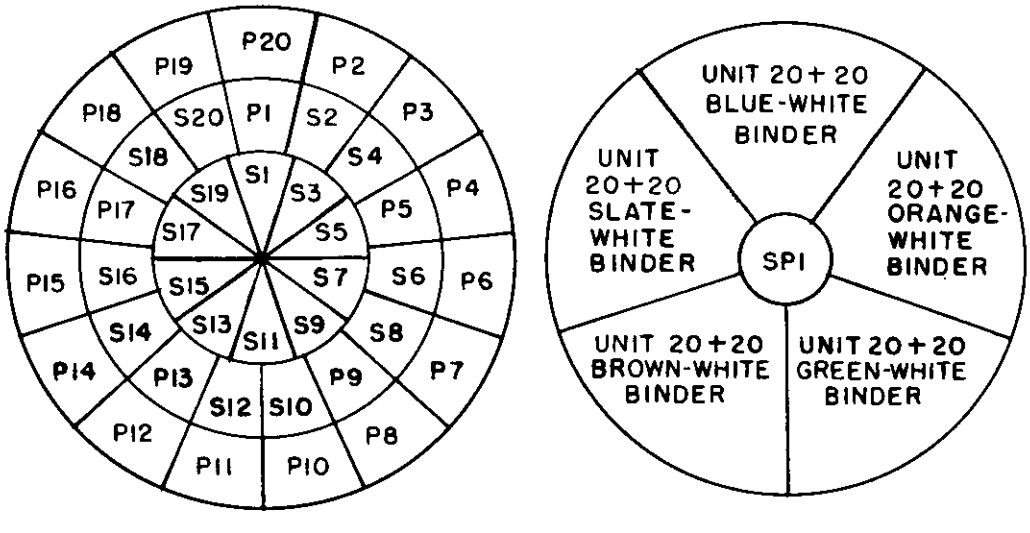
20 PAIR UNIT



UNIT 20

302 CONDUCTORS (100 PAIRS - 100 SINGLES, WITH 1 SPARE PAIR)

20 P 20 S UNIT



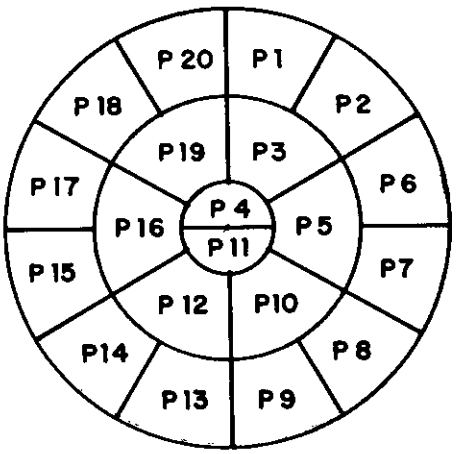
UNIT 20+20

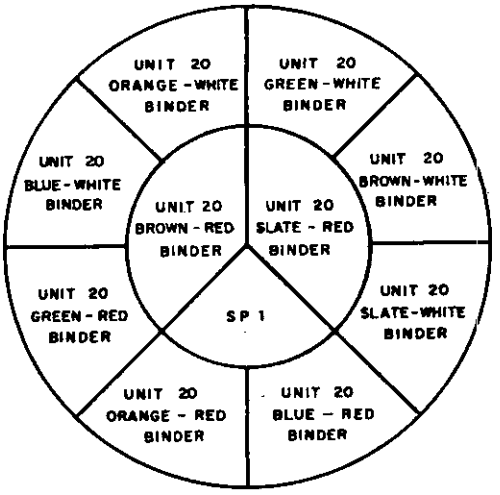
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

FIGURE 6. CABLE CONFIGURATION - UNIT TYPE CONSTRUCTION

402 CONDUCTORS (200 PAIRS WITH ONE SPARE PAIR)

20 PAIR UNIT





UNIT 20

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

3.6.1 Binders. Each individual unit of unit constructed cables shall be spirally bound with a color coded binder of nylon yarn servings or polyethylene tape. The color code of the binders shall conform to the binder colors specified in the applicable cable configuration of Figures 1 through 5.

3.6.2 Core wrapping. The core shall be wrapped with an overlapping, spirally applied Kraft electrical grade paper tape having a 0.005 inch nominal thickness or with a polyester tape having a 0.002 inch nominal thickness. If a binder of polyethylene tape is used (see 3.6.1), the core wrapping may be omitted.

3.7 Rip cord. A rip cord of double-ply nylon yarn shall be laid over the wrapped cable core directly beneath the cable sheath.

3.8 Jacket. The outer covering shall be a slate colored plasticized, polyvinyl chloride or its copolymer with polyvinyl acetate in accordance with Specification MIL-I-3930, type JP. When the requirements of MIL-I-3930 are in conflict with 3.11 of this specification, the latter shall apply. The sheath shall be suitably stabilized, compounded, and extruded to provide the cable with a tough, flexible, long-life protective covering. The wall thickness of the plastic jacket shall be 0.030 inch minimum for core diameters less than 1 inch, and 0.040 inch minimum for core diameters 1 inch and over.

3.8.1 Diameter of finished cables. The outside diameter of the finished cable shall be in accordance with those specified in 1.2.

3.9 Electrical requirements.

3.9.1 Continuity. Each conductor in each cable shall be continuous, when tested as specified in 4.6.1.

3.9.2 Short circuits. Each conductor shall be free of any contact with any other conductor, when tested as specified in 4.6.2.

3.9.3 Resistance. The d.c. resistance of each conductor shall not exceed 94 ohms per mile of cable at 68°F, when tested as specified in 4.6.3.

3.9.4 Dielectric strength. The insulation of each conductor shall be capable of withstanding for one minute an AC potential of 1500 volts, when tested as specified in 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

3.9.5 Insulation resistance. The insulation resistance between each conductor and the remaining conductors connected together shall be not less than 4000 megohms-1000 feet at 68°F, when tested as specified in 4.6.5.

3.10 Physical requirements for insulation.

3.10.1 Tensile strength. The insulation compound shall have a tensile strength of 2500 pounds per square inch, when tested as specified in 4.7.1.

3.10.2 Elongation. The insulation compound shall have an elongation of 100 percent minimum, when tested as specified in 4.7.2.

3.10.3 Adhesion. The maximum pull required to remove a 1-1/4 inch length of insulation shall be 3 pounds, when tested as specified in 4.7.3.

3.10.4 Shrinkage. The maximum allowable insulation shrinkback shall not exceed 1/16 inch, when tested as specified in 4.7.4.

3.10.5 Cold bend. The insulation shall not develop any cracks visible to the unaided eye, when tested as specified in 4.7.5.

3.10.6 Flame retardant properties. The insulation on a conductor shall not continue to burn but shall be self-extinguishing within 5 seconds, when tested as specified in 4.7.6.

3.10.7 Compression strength. The insulation shall withstand a compression test of 1000 pounds, when tested as specified in 4.7.7.

3.11 Physical requirements for jacket.

3.11.1 Tensile strength. The jacket compound shall have a minimum tensile strength of 1800 pounds per square inch, when tested as specified in 4.8.1.

3.11.2 Elongation. The jacket compound shall have an elongation of 200 percent minimum, when tested as specified in 4.8.2.

3.11.3 Aged elongation. The jacket compound shall have a minimum aged elongation of 100 percent, when tested as specified in 4.8.3.

3.11.4 Heat shock. The jacket of a section of the completed cable shall show no cracks when the cable is tested, as specified in 4.8.4.

3.11.5 Cold bend. The jacket shall not develop any cracks visible to the unaided eye when tested, as specified in 4.8.5.

3.11.6 Flame resistance. The jacket shall not continue to burn but shall be self-extinguishing within 5 seconds, when tested as specified in 4.8.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

3.11.7 Jacket stripping. The cable jacket shall not adhere to the cable and shall be freely skinned from the cable with the rip cord, when tested as specified in 4.8.7.

3.12 Workmanship. Cables shall be constructed and finished in a thoroughly workmanlike manner in accordance with accepted high grade production techniques so that the final product will meet all applicable requirements of this specification and any referenced subsidiary documents when inspected in accordance with the visual and dimensional examination of 4.9. The cables shall be a uniform and consistent product and shall be free from any defects which will adversely affect the serviceability of the product, such as lumps, kinks, splits, abrasions, scrapes, corroded surfaces, skin impurities and faulty extruded surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Inspection covered by subsidiary documents (see 4.3).

(b) Quality conformance inspection.

(1) Quality conformance inspection of cables before preparation of delivery. (See 4.4.)

(2) Quality conformance inspection of preparation for delivery. (See 4.5.)

4.3 Inspection covered by subsidiary documents. The materials listed in table III shall be inspected under the applicable subsidiary documents as part of the inspection of cables before preparation for delivery.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

Table III. Subsidiary documents inspection.

Materials	Requirement paragraph	Inspection document	AQL
Bare conductors Insulation compound Jacketing compound	3.4 3.5, 3.10 3.8, 3.11	QQ-W-343 MIL-I-3930 MIL-I-3930	Per applicable specification unless otherwise specified herein.

4.4 Quality conformance inspection of cables before preparation for delivery. The contractor shall perform the inspection specified in 4.3 through 4.9. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedure and examine the contractor's inspection records. In addition the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.3.) Test equipment for Government verification inspection shall be made available by the contractor.

4.4.1 Group A inspection. This inspection, including sampling, shall conform to table IV and the inspection procedures of Standard MIL-STD-105, using general inspection level I. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table IV. Group A inspection.

Inspection	Requirement paragraph	Inspection paragraph	AQL Major	AQL Minor
Visual and dimensional	3.12	4.9	1.0% for the group combined	4.0% for the group combined
Construction	3.2			
Conductor size	3.4			
Joints	3.4.1			
Insulation thickness	3.5.1			
Diameter of insulated conductor	3.5.2			
Color code	3.5.3, 3.5.3.1, 3.5.3.2			
Cabling	3.6, 3.6.1			
Core wrapping	3.6.2			
Rip cord	3.7			
Jacket thickness	3.8			
Diameter of finished cables	3.8.1			
Electrical				
Continuity	3.9.1	4.6.1	1.0% for the group combined	*
Short circuit	3.9.2	4.6.2		
Resistance	3.9.3	4.6.3		
Dielectric strength	3.9.4	4.6.4		
Insulation resistance	3.9.5	4.6.5		

*All electrical defects are considered major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

4.4.2 Group B inspection. This inspection, including sampling, shall conform to table V and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. The AQL shall be 4.0 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall normally be performed on inspection lots that have been subjected to and met the Group A inspection.

4.4.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

Table V. Group B inspection

Inspection	Requirement paragraph	Inspection paragraph
Insulation		
Tensile strength	3.10.1	4.7.1
Elongation	3.10.2	4.7.2
Adhesion	3.10.3	4.7.3
Shrinkage	3.10.4	4.7.4
Cold bend	3.10.5	4.7.5
Jacket		
Tensile strength	3.11.1	4.8.1
Elongation	3.11.2	4.8.2
Heat shock	3.11.4	4.8.4
Jacket stripping	3.11.7	4.8.7

4.4.3 Group C inspection. This inspection shall be as listed in table VI and shall normally be performed on sample units that have been subjected to and met Group A and B inspection.

Table VI. Group C inspection.

Inspection	Requirement paragraph	Inspection paragraph
Insulation		
Flame test	3.10.6	4.7.6
Compression	3.10.7	4.7.7
Jacket		
Flame test	3.11.6	4.8.6
Aged elongation	3.11.3	4.8.3
Cold bend	3.11.5	4.8.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

4.4.3.1 Sampling procedure. Three sample units shall be selected each month, without regard to their quality, and sufficient specimens shall be cut from each sample unit to perform the Group C inspection.

4.4.3.2 Noncompliance. If a sample fails Group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.5).

4.5 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with specification MIL-P-116 to determine conformance to the requirements of section 5.

4.6 Electrical inspection.

4.6.1 Continuity inspection. Test all conductors with a low d.c. voltage and an appropriate indicating meter to determine conformance with 3.9.1.

4.6.2 Short circuit inspection. Test each conductor individually to determine conformance with 3.9.2. Each conductor shall be free of any contact with any of the other conductors which may be bunched together to expedite testing. Use a test voltage of 100 volts DC, minimum.

4.6.3 Resistance inspection. Measure the resistance of the conductors to determine that it does not exceed the 94 ohms at 68°F specified in 3.9.3.

4.6.4 Dielectric strength inspection. Apply 1500 volts AC between each conductor in turn and the remaining conductors connected together for one minute to determine conformance with 3.9.4. The conductors may be split into groups so that neither adjacent single conductors nor both conductors of a pair are in the same group.

4.6.5 Insulation resistance inspection. Following the dielectric strength test, determine the insulation resistance between each conductor in turn and the remaining conductors connected together at 68°F. Use a testing voltage of not less than 100 volts DC nor more than 550 volts DC. The determined value shall be not less than 4000 megohms-1000 feet as required in 3.9.5.

4.7 Physical inspection for insulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

4.7.1 Tensile strength inspection. The insulation compound shall be tested in accordance with Method 3021 of Federal Test Method Std No. 228, to determine conformance with 3.10.1.

4.7.2 Elongation inspection. The insulation compound shall be tested in accordance with Method 3031 of Federal Test Method Std No. 228, to determine conformance with 3.10.2.

4.7.3 Adhesion inspection. Determine the insulation adhesion by removing 4-1/2 inches from one end and 1/4 inch from the other end of a 6 inch sample of insulated conductor. Place the 4-1/2 end of the conductor through a 1/8 inch thick brass plate having a hole approximately 0.010 inch larger than the diameter of the wire. Make a loop in the long end of the sample and attach a tension testing scale to this loop. Pull the scale at the rate of approximately 10 feet per minute to determine that the maximum pull required to remove the conductor from the 1-1/4 inch of insulation is not over 3 pounds, as required in 3.10.3.

4.7.4 Shrinkage inspection. Skin 1/2 inch on one end of a 6 inch sample of insulated conductor taken from the finished cable. Immerse the skinned end 1/4 inch deep for 20 seconds into a pot of solder at approximately 320°C. The shrinkback of the insulation shall not exceed 1/16 inch, as specified in 3.10.4.

4.7.5 Cold bend inspection. Condition an 8 inch sample of insulated conductor and a 1/16 inch mandrel for one hour at -18°C(0°F). At the end of the hour while the sample and mandrel are still in the cold box, wrap the sample five consecutive turns in a close helix about the mandrel at the rate of approximately 60 turns per minute. Examine the insulation to determine compliance with 3.10.5.

4.7.6 Flame retardant inspection. Suspend an 18 inch length of insulated conductor vertically in a draft free inclosure and apply the flame of a Bunson burner for 5 seconds. The flame shall be applied so that the apex of the blue cone is played on the lower portions of the wire at an angle of approximately 45 degrees. After removing the flame the insulation shall extinguish itself in 5 seconds, as required in 3.10.6.

4.7.7 Compression strength inspection. Condition the test specimen of insulated wire for at least one hour at a temperature of 25°C ± 3°C at which temperature the compression strength shall be determined. Place the test specimen between 2 inch by 2 inch plates and measure the crushing load required to cause contact between the wire and either of the plates when the approach speed of the plates is approximately 2-1/4 inches per minute. The insulation shall withstand a 1200 pound compression load without failing, as required in 3.10.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

4.8 Mechanical inspection of jacket.

4.8.1 Tensile strength inspection. The jacket compound shall be tested in accordance with method 3021 of Federal Test Method Std. No. 228 to determine conformance with 3.11.1.

4.8.2 Elongation inspection. The jacket compound shall be tested in accordance with Method 3031 of Federal Test Method Std. No. 228 to determine conformance with 3.11.2.

4.8.3 Aged elongation inspection. Oven-age the jacket compound for 48 hours at 120°C ± 2°C and then test in accordance with Method 4001 of Federal Test Method Std. No. 228 to determine conformance with 3.11.3.

4.8.4 Heat shock inspection. Wind six close turns of completed cable around a mandrel and fasten the ends securely to the mandrel. The diameter of the mandrel shall be three times the overall diameter of the cable for all cables under 3/4 inch. For cables 3/4 inch and larger, the mandrel diameter shall be eight times the overall diameter of the cable. Place the cable on mandrel in an oven maintained at 120°C ± 2°C for a period of one hour. Remove from the oven and allow to cool to room temperature. Remove the cable from the mandrel and straighten. Slit the jacket lengthwise and remove from the core. Examine the sheath to determine conformance with 3.11.4.

4.8.5 Cold bend inspection. Condition a sample of completed cable in a cold box at -18°C ± 2°C for 24 hours. After the 24 hours of exposure, remove the sample from the cold box and immediately bend it 180 degrees around a mandrel of the size specified below at a uniform rate of bend. Complete the bending within 10 seconds. Examine the jacket to determine compliance with 3.11.5.

Number of Conductors	Mandrel Diameter (Inches)
8 - 13	1/2
14 - 24	1-1/2
25 - 65	2
66 - 109	2-1/2
110 - 205	4
206 - 305	8
306 - 405	12

4.8.6 Flame resistance inspection. Expose a sample of the jacket to a Bunsen burner flame for 15 seconds in a draft free area. Remove the Bunson flame and determine whether the jacket flame goes out within 5 seconds Repeat three times at 15 second intervals to determine that the jacket is self-extinguishing and does not continue to burn, as required by 3.11.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

4.8.7 Jacket stripping inspection. Pull the rip cord to cut a section of the jacket. Remove the jacket to determine that it skins freely and clearly, as required by 3.11.7.

4.9 Visual and dimensional inspection. Inspect the completed cables to verify that the materials and all workmanship comply with this specification. At the option of the Government, certification as to materials and splices may be accepted.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking shall be in accordance with the applicable provisions of MIL-C-12000. (See 6.2(c).)

6. NOTES

6.1 Intended use. The switchboard cable covered by this specification is intended for use indoors for interconnecting multiple appearances of circuits on switchboards with the distibuting frame.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type and designation of cable to be furnished (see 1.2).

(c) Level of packaging and level of packing required for shipment (see Section 5).

6.3 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance requirements established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.4 Definitions.

6.4.1 For the purpose of this specification, the definitions given below shall apply:

6.4.1.1 Lot size. The lot size is the number of units of product in an inspection lot. The minimum and maximum lot sizes for inspection purposes shall be determined by the Government, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purposes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

6.4.1.2 Unit of product. The unit of product is a continuous length of finished cable contained on a reel.

6.4.1.3 Sample. A sample is one or more sample units selected from an inspection lot to represent that inspection lot for inspection purposes.

6.4.1.4 Sample unit. A sample unit is a unit of product, selected in an unbiased manner to be part of a sample, from which specimens are taken for inspection.

6.4.1.5 Specimen. A specimen is a single piece of finished cable which is taken from a sample unit and subjected to inspection.

6.4.1.6 Finished cable. Finished cable is cable on which all manufacturing operations have been completed and which is ready to be submitted for acceptance.

6.5 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3.2).

6.6 Revision changes. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - EL
Air Force - 80

Review:
Army
Air Force - 80
NSA
DSA - IS

Users:
Army - MI, ME
Air Force - 17

Preparing Activity:
Army - EL

Code "C"

Project Nr. 6145-0526

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
HEADQUARTERS
U.S.ARMY ELECTRONICS COMMAND
FORT MONMOUTH, NEW JERSEY 07703

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding General
U. S. Army Electronics Command
ATTN: AMSEL- TD-S
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

GPO 303-224/Z 896

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification *which will insure that suitable products* can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-C-55446A Cables, Telephone, Switchboard:Plastic Insulated,Plastic Jacketed

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:22Z
Check the source to verify that this is the current version before use.